Arnav Chaturvedi

arnavc.work@gmail.com | Mumbai, India | linkedin.com/in/arnavchat | github.com/Arnav-3012

EDUCATION

NMIMS University (MPSTME) July 2022 - May 2028 (Expected)
BTech in Data Science | CGPA: 3.83/4 Mumbai, India

EXPERIENCE

AI Developer Oct 2026 - Present
Credit Intelligence (UAE fintech startup), Remote

- Improve retrieval on a production RAG assistant for credit card comparison (FastAPI, pgvector, LangChain).
- Harden LLM data extraction with Pydantic validation and self-correcting retries, shipped via CI/CD with 300+ tests.

AI / Data Engineering Intern June 2026 - July 2026
Ticker Ltd, Mumbai, India

- Automated Grafana reporting for 3 teams (Python, Streamlit, Playwright, SMTP), saving ~6 hours of manual work a week.
- Built a KOL mapping tool (Streamlit, MySQL procedures, audit logs) giving 10+ business users safe write access.
- Migrated 40+ tables from MySQL to PostgreSQL with integrity checks; wrote the team's migration onboarding guide.

AI Solutions Intern July 2025 - October 2025
Wingreens World (FMCG), Mumbai, India

- Built AI sales enablement tools and data pipelines across 4 FMCG brands; prototypes joined the AI roadmap.
- Shipped an LLM chatbot and dashboards with stakeholders, cutting repeat data requests to the ops team by ~30%.

Project Trainee, AI / Data Science May 2025 - June 2025
Datamatics Global Services Ltd., Mumbai, India

- Auto-assigned GL codes for 500+ vendors via top-N embedding retrieval (~86% accuracy), deployed to production.
- Extracted and matched company names and IDs from raw bank statements with NLP, cutting reconciliation work by ~40%.
- Rewrote a recurring Excel workflow in Python (pandas), cutting runtime from 15+ minutes to under 30 seconds.

Co-Founder & CEO Feb 2024 - Feb 2025
Absolute Wear (D2C clothing startup), Mumbai, India

- Built a D2C clothing brand from scratch to 100+ customers in 45 days, owning design, production and marketing.

PROJECTS

AgentGate: Cost-Aware Adaptive RAG Router Python, LangGraph, Qdrant, AWS

- Routed queries between single-pass RAG and a capped agent loop, cutting cost ~40% on a 500-query benchmark.
- Wrote the agent engine (orchestration, tools, memory, stopping rules) with live cost and latency tracking on AWS.

Adaptive Noise Cancellation for Defence Communication PyTorch, ONNX, TFLite, CoreML

- Fine-tuned a speech enhancement model on LibriSpeech and MUSAN to strip rotor, siren and gunfire noise live.
- INT8-quantized for TFLite/CoreML; benchmarked vs an STFT baseline on SNR >15 dB, STOI >0.85, PESQ >2.5 targets.

FireWatch: Fire Detection and Alert System PyTorch, ONNX, LangGraph, AWS Lambda

- Trained MobileNetV3 on 18.7K images: 95.75% fire recall, 86.6% precision, 91.3% accuracy; deployed via ONNX.
- Built a LangGraph alert agent that fuses camera and MQ-2/MQ-135 gas signals into WATCH, WARNING and CRITICAL levels, then sends Telegram/SMS alerts with the nearest fire stations via the Overpass API.
- Added an AWS Lambda second opinion, cut bright-wall false smoke alerts from 98% to 0%, passed 11/12 live trials.

StatCalib: Medical AI Calibration Audit PyTorch, SciPy, Streamlit

- Audited DenseNet-121 calibration on 112K NIH X-rays via Hosmer-Lemeshow tests; Platt scaling cut ECE by 93.5%.

Hospital Operations Intelligence System PostgreSQL, FastAPI, LLM, JWT

- Built a 12-table PostgreSQL backend with 63 FastAPI endpoints and LLM NL-to-SQL, ~85% correct on 50 test questions.

SKILLS

ML and Languages: Python, SQL, PyTorch, scikit-learn, pandas, Machine Learning, Deep Learning, Computer Vision, NLP

Generative AI: LLMs, Retrieval-Augmented Generation (RAG), AI Agents, LangChain, LangGraph, Vector Databases (Qdrant, pgvector), Transformers, Prompt Engineering, LLM Evaluation

Backend and Data: FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Pydantic, Data Pipelines, ETL, Data Quality

Deployment and Cloud: AWS (Lambda, DynamoDB, CloudWatch), Model Deployment (ONNX, TFLite), Quantization, CI/CD, Git

REFERENCES

- Zoha Chaudhri, Co-founder and AI Systems & Innovation Lead, Wingreens World | Letter of Recommendation